Roll No. ....

TOE–805

B. TECH. (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

ROBOTICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Explain the force and torque control of manipulators. (CO2, CO5)

   (b) Justify the statement : Actuators are the muscles of robots. (CO2, CO5)

   (c) Discuss the polar coordinate robot configuration. Also, discuss in brief about Pneumatic drives. (CO2, CO5)

2. (a) What is path planning in the context of robotic automation ?(CO4, CO2)

   (b) Discuss the different characteristics of sensors. (CO4, CO2)

   (c) Give a short explanation on the static force and moment analysis of industrial robots. (CO4, CO2)

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3. (a) Robots find applications not only in the industry. Explain three non industrial applications of robots. (CO1, CO3)

   (b) What are the advantages and disadvantages of electric drive systems ? (CO1, CO3)

   (c) Discuss the basic objectives of Robotics and discuss the point of differences between hydraulic and electric drives. (CO1, CO3)

4. (a) Discuss the Industrial robot anatomy. (CO3, CO2)

   (b) What is path planning in the context of robotic automation ?(CO3, CO2)

   (c) Discuss Range sensors and proximity sensors. And list out their differences. (CO3, CO2)

5. (a) Provide the typical features of robotic actuators and drive systems. (CO4, CO5)

   (b) Describe any three applications of image processing in robotics. (CO4, CO5)

   (c) Explain about joint angle, joint distance, link length and link twist. How link parameter table is obtained if you know these parameters ? (CO4, CO5)

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TME-805

B. TECH. (MECHANICAL) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

ENTREPRENEURSHIP ESSENTIALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain different types of entrepreneurships. (CO1, CO2)

   (b) Why do we need deliberate practice in entrepreneurship ? Explain the our steps of the same. (CO1, CO2)

   (c) What is entrepreneurial mindset ? Explain. (CO1, CO2)

2. (a) Compare growth and fixed mindset with five points. (CO1, CO3)

   (b) Explain innovation, Invention, Improvement and irrelevant with one example each. (CO1, CO3)

   (c) Enlist the seven strategies of idea generation. (CO1, CO3)

P. T. O.

3. (a) What are the four pathways of opportunity identification ? Explain.

(CO1, CO2 & CO3)

   (b) Explain disruptive business model with two examples.

(CO1, CO2 & CO3)

   (c) Enlist the most common components of business model.

(CO1, CO2 & CO3)

4. (a) Explain five elements of design thinking. (CO1, CO2)

   (b) Enlist *five* different methods of ideation and explain any two of them with example. (CO1, CO2)

   (c) What is business model canvas ? Explain with an example and diagram.

(CO1, CO2)

5. (a) What is power and value of networks in testing of new ideas for entrepreneurship ? (CO4, CO2)

   (b) What is bootstrapping. Explain with an example. (CO4, CO2)

   (c) Differentiate between angel investor and venture capitalist with *five* points. (CO4, CO2)

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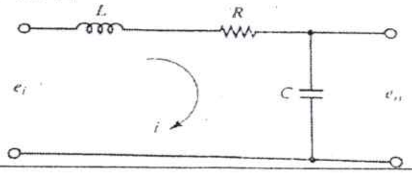
EVEN END SEMESTER EXAMINATION - 2024

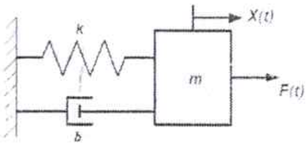
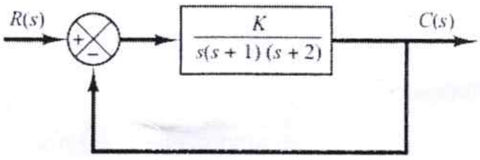
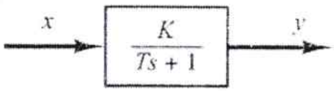
Name of the Course: B.Tech. (ME) — *Semester:* **VIII**

Name of the Paper: Automatic Control — *Paper Code:* **TRA 801**

Time: 3 Hours — Maximum Marks: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	**Describe the open-loop and closed-loop systems with neat block diagrams? Also, discuss the disadvantages of a closed-loop system.**	CO1
(b)	**Find the solution x(t) of the following differential equation:** $\ddot{x} + 2\dot{x} + 5x = 3\ ;\ \ where\ \ x(0) = 0\ \ and\ \ \dot{x}(0) = 0$	CO1
(c)	**Plot and write the expressions (with Laplace transform) for the following signals: step, ramp, sinusoidal, and impulse.**	CO1
Q2	**(20 marks)**	
(a)	**What is the characteristic equation of a system? Describe the following system responses: undamped, overdamped, underdamped, and critically damped systems in terms of roots of the characteristic equation.**	CO2
(b)	**Linearize the following nonlinear equation around an operating point of x=2.** $y = 0.2x^3$ **Also, check the error, if the operating point is changed to x=3.**	CO2
(c)	**Derive the differential equations and transfer function of the following electrical R-L-C circuit. Consider input is e_i and output is e_o. Also, discuss about the stability of the system's response, when L=0.2 H, R=0.5 Ohm, and C=0.1 F.** 	CO2
Q3	**(20 marks)**	
(a)	**Derive the expression for the unit-step response of a first-order system.**	CO3
(b)	**Comment about the stability of a system using Routh's stability criterion, whose characteristic equation is given by the following polynomial:** $s^4 + s^3 + 4s^2 + 2s + 7 = 0$	CO3

(c)	Describe the following transient-response characteristics of a control system: (i) Delay time (ii) Rise time (iii) Peak time (iv) Maximum overshoot and (v) Settling time.	
Q4	(20 marks)	
(a)	What do you understand by the modern control theory? Write the standard state space equations for a linear and nonlinear dynamic system.	CO4
(b)	Derive the state space model of the mass-spring-dashpot system shown in the figure below. 	
(c)	Plot the root locus of the following closed-loop system. 	
Q5	(20 marks)	
(a)	Describe the Proportional-Integral-Derivative (PID) control action with governing equations.	CO5 & CO6
(b)	Obtain the steady state output y of the system shown in the figure below for a given sinusoidal input $x = Asin(\omega t)$.  $G(s)$	
(c)	Draw Bode plots for the integral term G(s)=K/s.	

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DEMM–08

B. TECH. (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

ADVANCED WELDING TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss about the safety aspect of various welding processes. List various safety equipment and accessories used during welding processes. (CO1)

(b) Differentiate between TIG and MIG welding process with their working principle, advantages, disadvantages and fields of application. (CO1)

(c) Explain principle of operation of resistance seam welding process with its advantages and disadvantages. (CO1)

P. T. O.

2. (a) Describe principle of operation of electron beam welding. What are the possible problems or difficulties and how it can be dealt with ? Write down the advantage and limitation. (CO2)

   (b) Discuss about the welding technique used in underwater welding. (CO2)

   (c) Differentiate between Oxy-acetylene gas cutting and Electric arc cutting with their advantages and disadvantages. (CO2)

3. (a) Define weldability. Explain effects of alloying elements on weldability. What is the purpose of weldability test ? Explain hot cracking test with neat sketch. (CO3)

   (b) Describe the main metallurgical effects of welding on materials. (CO3)

   (c) Explain the effect of carbon and alloying elements on weldability of steel. (CO3)

4. (a) Briefly explain the heat-affected-zone (HAZ) and its effects on welding properties. (CO4)

   (b) Describe the working principle of arc fusion welding with neat sketch. (CO4)

   (c) What is arc blow ? How you can prevent the arc blow ? (CO4)

5. (a) Explain magnetic particle inspection method with neat sketch. (CO5)

   (b) Explain ultrasonic inspection method for welding with neat sketch. (CO5)

   (c) List various non destructive testing methods and explain radiographic inspection method for weld joint. (CO5)

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DEEM–06

B. TECH. (ME) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

TOTAL QUALITY MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the Juran's economic Model for quality costs in detail. (CO1)

   (b) Explain the concept Six Sigma implementation in any organization; Also write the importance of six sigma. (CO1, CO2)

   (c) What is product ? Describe the double diamond product design process in detail. What is the difference between model and prototype of the product ? (CO2, CO3)

2. (a) Explain Juran's quality Trilogy in detail. Aiso Explain the relevance of six Sigma concept in achieving quality output. in the process. (CO3, CO4)

   (b) Design customer satisfaction questionnaire to evaluate the level of the customer satisfaction in sport shoe manufacturer industry. (CO4)

   (c) Explain the various types of costs contributing to the cost of quality. Give examples of each. (CO5)

3. (a) Write short notes on the following : (CO3)

   (i) Pareto analysis

   (ii) Check sheets

(iii) Histogram

(b) Ten samples of equal size are taken to prepare a P-chart .The total number of parts in these ten samples was 1200 and total number of defects counted was 155 determine the UCL, LCL for the P-chart. (CO4)

(c) Write the step by step procedure of implementing a bench marking process in an organization. (CO4, CO5)

4. (a) What is lean production system ? Describe in details Just in Time manufacturing system. (CO4)

(b) What are the elements of ISO 9000 quality system explain how quality systems are implemented documented and audited ? (CO3, CO4)

(c) What is Total Quality Management and its functions ? Also write the pillars of TQM ? (CO4)

5. (a) What is internal quality audit and external quality audit ? Also explained the role of senior management commitment in the implementation of quality system. (CO5, CO6)

(b) A welding workshop receives a lot of complaints about poor quality solder joints. Analyze this problem by constructing a cause and effect diagram. (CO5, CO6)

(c) With Suitable example explain various stages of building a House of quality matrix. (CO5, CO6)

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TAE–801

B. TECH. (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

VEHICLE BODY SAFETY AND CONTROL

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the key differences between active and passive safety systems in automobiles, and how do they contribute to overall vehicle safety ? (CO3 & CO4)

(b) Explain the purpose of crumple zones in vehicle design. How do crumple zones help in reducing the impact force experienced by occupants during a collision ? (CO3 & CO4)

(c) Describe the different types of crash tests used to evaluate vehicle safety, such as movable barrier tests and rollover crash tests. What are the main objectives of these tests ? (CO3 & CO4)

2. (a) Discuss the importance of structural safety and passenger compartment integrity in crash testing. How does this aspect of vehicle design contribute to occupant safety during a collision ? (CO2 & CO5)

(b) What are some common strategies employed to protect pedestrian from serious injury in the event of a vehicle-pedestrian collision ? (CO2 & CO5)

(c) How do systems like Electronic Power Assisted Steering (EPAS) and Automatic Braking contribute to the active safety of a vehicle ? (CO2 & CO5)

3. (a) Explain the role of restraint systems, such as seat belts and airbags, in protecting occupants during a crash. How do these systems won together to enhance safety ? (CO5 & CO6)

(b) Discuss the importance of bumpers in automobiles. What are the key design considerations for bumpers to ensure they provide effective impact protection ? (CO2 & CO5)

(c) What are the different types of safety glass used in automobiles, and why is safety glass preferred over traditional glass for vehicle windows and windshields ? (CO2 & CO5)

4. (a) Why is ergonomics important in automotive safety, and how does I impact the design and location of vehicle controls ? (CO4 & CO5)

(b) What are the key factors that determine human impact tolerance in rash, and how do these factors inform the development of injury criteria and safety thresholds ? (CO2 & CO5)

(c) How does anthropometry play a role in the design of vehicle interior to ensure safety and comfort for occupants ? (CO2 & CO5)

5. (a) Explain the function of cruise control systems in vehicles. How do digital cruise control systems differ from traditional mechanical systems ? (CO1 & CO2)

(b) What is an Antilock Braking System (ABS), and how does it contribute to safer vehicle operation, especially in emergency braking situations ? (CO2 & CO5)

(c) Describe the purpose of tire slip controllers in modern vehicles. by Jo these systems help maintain vehicle stability during chal1engin driving conditions ? (CO2 & CO5)

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TAS–801

B. TECH. (MECHANICAL ENGINEERGIN)
(EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

AIRPORT AND AIRLINE MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is SWOT analysis for airline company ? (CO1)

   (b) What are the 4 types of threats and challenges in the aviation industry ? (CO2)

   (c) What are the threats to low cost airlines ? (CO1)

2. (a) What does DGCA Stands for and what are its roles and functions ? (CO2)

   (b) How important is airport infrastructure in the Indian economy ? (CO2)

P. T. O.

(c) What are the main functions and responsibilities of Airports Authority of India ? (CO2)

3. (a) How do recent global developments influence the attractiveness of Public- Private Participation (PPP) models for Indian airports amidst increasingly stringent environmental regulations ? (CO3)

   (b) What strategies can Indian airports use to adapt to international trends in PPP amidst environmental regulations ? (CO3)

   (c) What types of modern techniques are used in controlling the security problems ? (CO3)

4. (a) Why is the human factor crucial in aircraft maintenance ? (CO4)

   (b) What human factors categories can lead to aircraft accidents or incidents ? (CO4)

   (c) Describe the diverse steps undertaken by various security personnel to conduct security checks on a passenger departing from one Airport to other, including baggage inspection. (CO6)

5. (a) How does Air Traffic Control facilitate the safe and efficient movement of aircraft within the airspace, and what are its key functions in airport operations ? (CO5)

   (b) Describe the protocols and procedures for handling unaccompanied minors at airports, including check-in, boarding, and supervision throughout their journey. (CO5)

   (c) Briefly explain the process of Cargo Handling. (CO6)